Chapter 8

Alcohols, Phenols and Ethers

Alcohols

1. Given below are two statements: *(2022)*

Statement I: In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. HCl + $ZnCl_2$, known as Lucas Reagent.

Statement II: Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

In the light of the above statements, choose the most appropriate answer from the options given below:

a. Statement I is incorrect but Statement II is correct.

b. Both Statement I and Statement II are correct.

c. Both Statement I and Statement II are incorrect.

d. Statement I is correct but Statement II is incorrect.

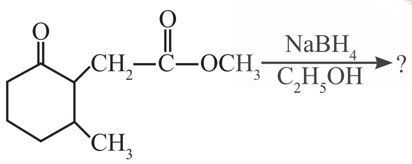
2. What is the IUPAC name of the organic compound formed in the following chemical reaction? *(2021)*

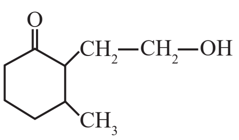
$$\text{Acetone} \xrightarrow[\text{(ii) } H_2O,\ H^+]{\text{(i)} C_2H_5MgBr,\ \text{dry ether}} \text{Product}$$

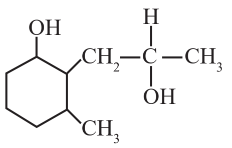
a. Pentan-2-ol   b. Pentan-3-ol

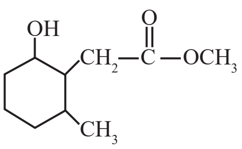
c. 2-methyl butan-2-ol   c. 2-methyl propan-2-ol

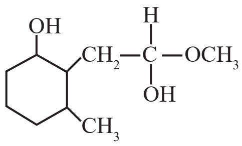
3. The product formed in the following chemical reaction is: *(2021)*



a. 

b. 

c. 

d. 

4. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give : *(2020)*

a. Sec. butyl alcohol   b. Tert. butyl alcohol

c. Isobutyl alcohol   d. Isopropyl alcohol

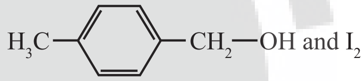
5. $CH_3CH_2CH=CH_2 \xrightarrow[H_2O,\ H_2O/OH]{B_2H_6} Z$

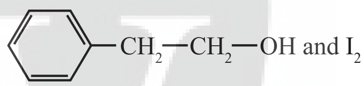
What is Z? *(2020-Covid)*

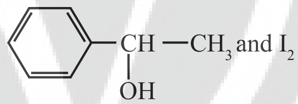
a. $CH_3CH_2\underset{\underset{OH}{|}}{C}HCH_3$ b. $CH_3CH_2CH_2CHO$

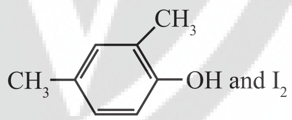
c. $CH_3CH_2CH_2CH_3$ d. $CH_3CH_2CH_2CH_2OH$

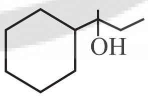
6. Compound A, $C_8H_{10}O$, is found to react with NaOI (produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell. A and Y are respectively *(2018)*

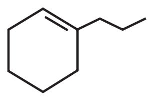
a. 

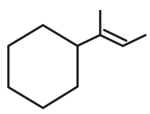
b. 

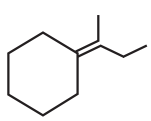
c. 

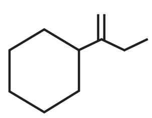
d. 

7. Which of the following is not the product of dehydration of  ? *(2015 Re)*

a. 

b. 

c. 

d. 

Phenols

8. Given below are two statements: *(2022)*

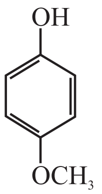
Statement I: The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrowing nitro group.

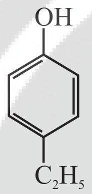
Statement II: o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

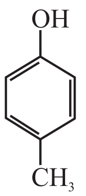
In the light of the above statements, choose the most appropriate answer form the options given below:

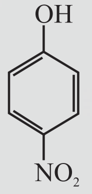
a. Statement I is incorrect but Statement II is correct.
b. Both Statement I and Statement II are correct.
c. Both Statement I and Statement II are incorrect.
d. Statement I is correct but Statement II is incorrect.

9. Which of the following substituted phenols is the strongest acid? *(2020-Covid)*

a. 

b. 

c. 

d. 

10. The compound that is most difficult to protonate is: *(2019)*

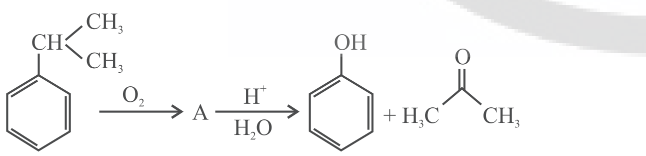
a. $H-O-H$

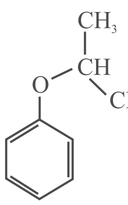
b. H_3C-O-H

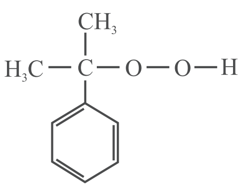
c. $H_3C-O-CH_3$

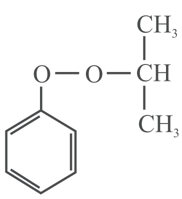
d. $Ph-O-H$

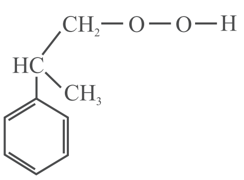
11. The structure of intermediate A in the following reaction, is *(2019)*



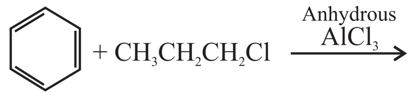
a. 

b. 

c. 

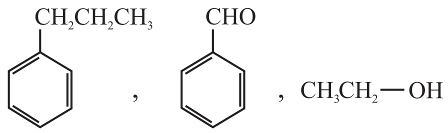
d. 

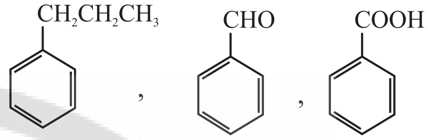
12. Identify the major products P, Q and R in the following sequence of reactions: *(2018)*

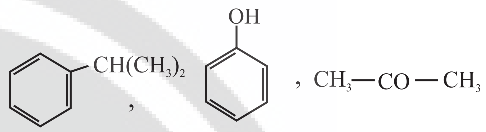


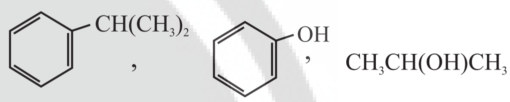
$$P \xrightarrow[\text{(ii) } H_3O/\Delta]{\text{(i)} O_2} Q + R$$

P Q R

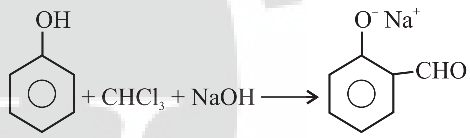
a. 

b. 

c. 

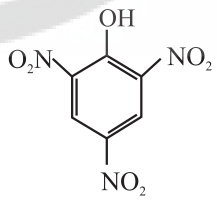
d. 

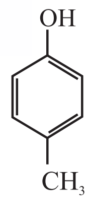
13. In the reaction, the electrophile involved is: *(2018)*

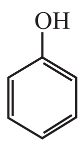


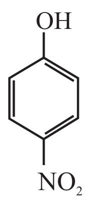
a. Dichloromethyl cation $\left(\overset{\oplus}{C}HCl_2\right)$
b. Formyl cation $\left(\overset{\oplus}{\text{ü}}\right)$
c. Dichlorocarbene $(:CCl_2)$
d. Dichloromethyl anion $\left(\overset{\ominus}{C}HCl_2\right)$

14. Which one is the most acidic compound? *(2017-Delhi)*

a. 

b. 

c. 

d. 

15. Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group? *(2015 Re)*

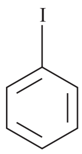
a. –CHO
b. $-CH_2Cl$
c. –COOH
d. $-CHCl_2$

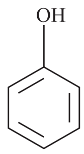
16. Which of the following will not be soluble in sodium hydrogen carbonate? *(2014)*

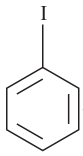
a. Benzoic acid
b. o-Nitrophenol
c. Benzenesulphonic acid
d. 2,4,6-trinitrophenol

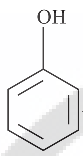
Ethers

17. Anisole on cleavage with HI gives: *(2020)*

a.  + CH_3OH

b.  + C_2H_5I

c.  + C_2H_5OH

d.  + CH_3I

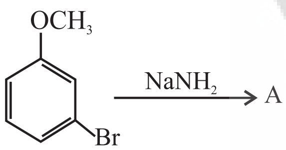
18. The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order: *(2018)*

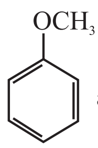
a. C_2H_5OH, C_2H_6, C_2H_5Cl
b. C_2H_5OH, C_2H_5Cl, C_2H_5ONa
c. C_2H_5OH, C_2H_5ONa, C_2H_5Cl
d. C_2H_5Cl, C_2H_6, C_2H_5OH

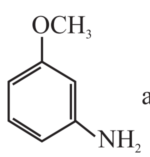
19. The heating of phenyl-methyl ethers with HI produces. *(2017-Delhi)*

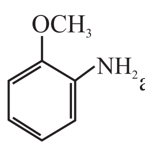
a. Benzene
b. Ethyl chlorides
c. Iodobenzene
d. Phenol

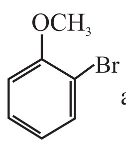
20. Identify A and predict the type of reaction: *(2017-Delhi)*



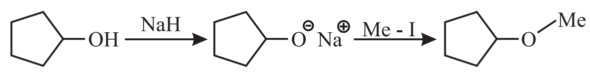
a.  and cine substitution reaction

b.  and substitution reaction

c.  and elimination addition reaction

d.  and cine substitution reaction

21. The reaction:



can be classified as? *(2016 - I)*

a. Williamson alcohol synthesis reaction
b. Williamson ether synthesis reaction
c. Alcohol formation reaction
d. Dehydration reaction

22. The reaction

$$\begin{array}{c} CH_3 \\ | \\ CH_3-C-ONa \\ | \\ CH_3 \end{array} + CH_3CH_2Cl \xrightarrow[-NaCl]{} \begin{array}{c} CH_3 \\ | \\ CH_3-C-O-CH_2-CH_3 \\ | \\ CH_3 \end{array}$$

is called: *(2015)*

a. Williamson continuous esterification process
b. Etard reaction
c. Gatterman - Koch reaction
d. Williamson synthesis

23. Among the following sets of reactants which one produces anisole? *(2014)*

a. C_6H_5OH; NaOH; CH_3I
b. C_6H_5OH; neutral $FeCl_3$
c. $C_6H_5-CH_3$; CH_3COCl; $AlCl_3$
d. CH_3CHO; RMgX

24. Identify Z in the sequence of reactions:

$$CH_3CH_2CH=CH_2 \xrightarrow{HBr/H_2O_2} Y \xrightarrow{C_2H_5ONa} Z$$ *(2014)*

a. $(CH_3)_2CH-O-CH_2CH_3$
b. $CH_3(CH_2)_4-O-CH_3$
c. $CH_3CH_2-CH(CH_3)-O-CH_2CH_3$
d. $CH_3-(CH_2)_3-O-CH_2CH_3$

25. Among the following ethers, which one will produce methyl alcohol on treatment with hot concentrated HI? *(2013)*

a. $CH_3-CH_2-CH_2-CH_2-O-CH_3$

b. $\begin{array}{l} CH_3-CH_2-CH-O-CH_3 \\ \qquad\qquad\quad | \\ \qquad\qquad\; CH_3 \end{array}$

c. $\begin{array}{c} CH_3 \\ | \\ CH_3-C-O-CH_3 \\ | \\ CH_3 \end{array}$

d. $\begin{array}{l} CH_3-CH-CH_2-O-CH_3 \\ \qquad\;\, | \\ \qquad CH_3 \end{array}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	c	c	b	d	c	a	d	d	d	b	c	c	a	a	b	d
18	19	20	21	22	23	24	25									
c	d	b	b	d	a	d	c									